



RAFFLES INSTITUTION
2015 YEAR 6 PRELIMINARY EXAMINATION
Higher 3

MATHEMATICS

9824

23 September 2015

3 hours

Additional materials: Answer Paper
Graph paper
List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

This document consists of 7 printed pages.

RAFFLES INSTITUTION
Mathematics Department

Answer **all** the questions.

1. Let $n \geq 2$ be a positive integer and $f_n(x) = x^n$ for $0 \leq x \leq 1$.

(a) Sketch the graphs of $y = f_n(x)$ and $y = f_{n+1}(x)$ on the same diagram. [1]

(b) (i) Sketch the graph of $y = \ln x$ and hence explain why $\frac{\ln a + m \ln b}{m+1} \leq \ln \left(\frac{a+mb}{m+1} \right)$ where a and b are positive real numbers and m is a positive integer. [2]

(ii) Hence obtain an inequality connecting $a+mb$ and ab^m . [2]

(iii) By writing $x^n(1-x)$ as $n \left(\frac{x}{n} \right)^n (1-x)$, find the maximum value of $x^n(1-x)$ and find the value of x for which the maximum is attained. [3]

Let a and b be real numbers such that $a < b$. The length L of the curve $y = f(x)$ joining the point $(a, f(a))$ and to the point $(b, f(b))$ can be evaluated by

$$L = \int_a^b \sqrt{1 + (f'(x))^2} \, dx.$$

Denote by L_n the length of the curve $y = f_n(x)$.

(c) (i) Show that for nonnegative real numbers m and n , $\sqrt{m+n} \leq \sqrt{m} + \sqrt{n}$ and determine when equality holds. [2]

(ii) Hence show that $L_n < 2$ for all positive integers n . [3]

2. A sequence of polynomials $P_n(x)$ is defined recursively by $P_1(x) = x - \frac{1}{2}$ and

$$\frac{d}{dx} P_n(x) = nP_{n-1}(x), \quad P_n(0) = P_n(1)$$

for $n \geq 2$.

- (i) By considering $P_3(0) = P_3(1)$, show that $P_2(x) = x^2 - x + \frac{1}{6}$. [2]

- (ii) Find $P_3(x)$ and $P_4(x)$. [3]

- (iii) Prove, using Mathematical Induction, that

$$P_{n+1}(x+1) = P_{n+1}(x) + (n+1)x^n$$

for $n \geq 0$. [4]

- (iv) Use the results in (ii) and (iii) to obtain a formula for $\sum_{x=0}^m x^3$. [3]

3. (i) Let a be a positive real number and f a function defined on $[0, a]$ satisfying $f(x) = f(a-x)$. Show that

$$\int_0^a xf(x) dx = \frac{a}{2} \int_0^a f(a-x) dx.$$

[2]

- (ii) Hence, prove that for $n \in \mathbf{Z}^+$,

$$\int_0^\pi x \sin^n x dx = \frac{\pi}{2} \int_0^\pi \sin^n x dx = \pi \int_0^{\frac{\pi}{2}} \sin^n x dx.$$

[2]

- (iii) For nonnegative integers n , let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$. Show that $nI_n = (n-1)I_{n-2}$ where $n \geq 2$. [3]

- (iv) Hence find the exact value of $\int_0^\pi x \sin^{11} x dx$. [3]

- (v) Show that the sequence defined by $x_n = nI_n I_{n-1}$ for $n \in \mathbf{Z}^+$ is constant. [1]

- (vi) Hence show that, for $n \in \mathbf{Z}^+$, $\sqrt{\frac{\pi}{2(n+1)}} \leq I_n \leq \sqrt{\frac{\pi}{2n}}$. [3]

4. (a) Harry has 42 identical sweets and he wants to give all of them to his six friends, Arry, Barry, Carrie, Denny, Elly and Fanny. Arry, Barry and Carrie say they want at least 3, 4 and 5 sweets respectively.

(i) What is the number of ways Harry can distribute the sweets to his friends? [2]

(ii) What is the number of ways Harry can distribute the sweets to his friends if, in addition, Denny, Elly and Fanny can take at most 8 sweets each? [4]

- (b) The Stirling number of the second kind, denoted by $S(r, n)$ is defined as the number of ways of distributing r distinct objects into n identical boxes such that no box is empty.

(i) Explain why $S(r, 2) = 2^{r-1} - 1$. [1]

(ii) Prove that $S(r, n) = S(r-1, n-1) + nS(r-1, n)$ for $r \geq n+1$. [2]

(iii) Find the number of ways Ginny can place her 5 distinct soft-toys into 3 identical bags, where some bags may be empty. [3]

5. A composition of n is where n is written as a sum of positive integers, and ordering of the summands matters. For example, 3 can be written using 4 compositions,

$$3 = 1+1+1 = 1+2 = 2+1.$$

- (a) Using a combinatorial argument, explain why the number of compositions for a positive integer n is given by

$$\sum_{k=1}^n \binom{n-1}{k-1}.$$

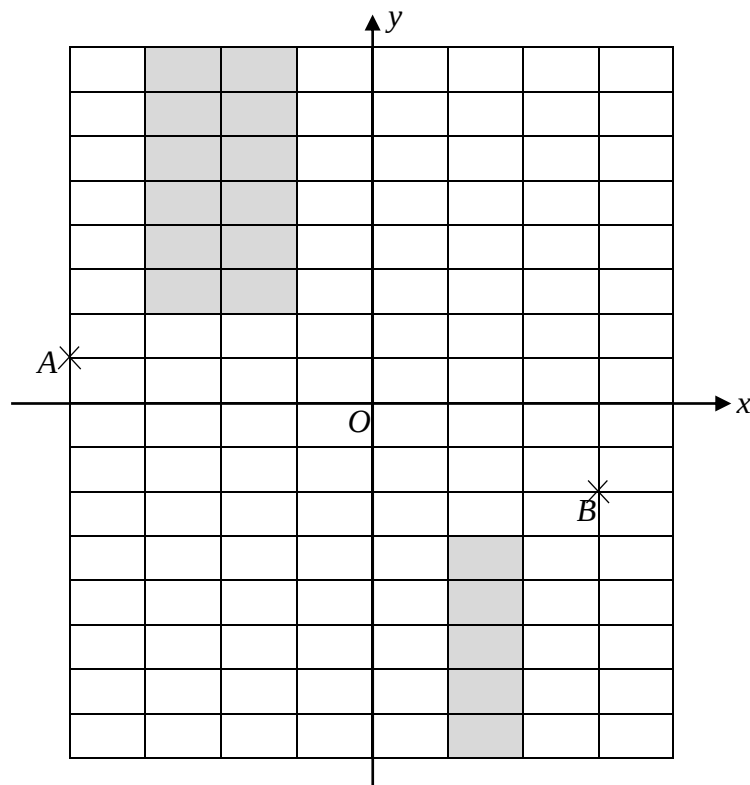
[2]

- (b) Let F_n be the number of compositions of $n+1$ using integers greater than 1. Using a combinatorial argument, show that for all positive integers n ,

(i) $F_n = \sum_{k=1}^{\lfloor \frac{n+1}{2} \rfloor} \binom{n-k}{k-1}$, where $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x . [2]

(ii) $F_{n+1} = F_n + F_{n-1}$. [2]

6. The diagram below shows a town with streets parallel to the x and y axes. The town hall is located at the origin O . Ana and Banana, two residents of the town, have their houses located respectively at $A(-4, 1)$ and $B(3, -2)$.



In this question, let $s(XY)$ denote the number of shortest routes from X to Y , moving along the streets of the town.

- (a) (i) Use a suitable bijection to show that $s(AB) = \binom{10}{3}$. [2]
- (ii) Find the number of shortest routes from Ana's house to Banana's house that do not pass through the town hall O . [1]
- (b) Ana and Banana decide to build new houses A' and B' respectively. They are located in the same quadrant as their respective original houses, but not along the x and y axes. How many ways are there to construct their new houses such that $s(A'B') = s(AB)$? [3]
- (c) Cena and Dena are new to the town and they decide to build their houses C and D in the same quadrant as A and B respectively, and on the vertices of the shaded regions. Show that for all possible choices of C and D ,

$$s(AB)s(CD) < s(AD)s(BC).$$

[2]

7. The motion of 2 particles A and B are each described by the differential equation

$$2 \frac{d^2 x}{dt^2} + 8 \frac{dx}{dt} + cx = 0.$$

Here, t denotes time, and x is the distance from a fixed point O on its line of motion. It is known that for particle A , $x = 2e^{-2t}$.

- (i) Show that $c = 8$. [1]
- (ii) State the type of damping both particles experience. [2]
- (iii) Particle B is initially at rest at $x = 3$. Find the distance of the particle from the fixed point O when $t = 3$, leaving your answer in exact form. [3]

8. (a) Find the general solution of the differential equation

$$\frac{dy}{dx} - \frac{2xy}{x^2 + 1} = 1. \quad [3]$$

- (b) Consider the second order differential equation

$$(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + p^2 y = 0, \quad (1)$$

where p is a positive integer.

- (i) Show that the substitution $x = \cos \theta$ reduces (1) to

$$\frac{d^2 y}{d\theta^2} + p^2 y = 0. \quad [4]$$

- (ii) Hence find the general solution to (1). [3]

- (iii) In the case $p = 3$, it is given that $y = 0$ and $\frac{dy}{dx} = -3$ when $x = 0$. Express this particular solution as a polynomial in x . [3]

9. The variables P and t are related by the “modified” logistic equation

$$\frac{dP}{dt} = \frac{1}{10} P \left(1 - \frac{P}{10} \right) (P - 1).$$

The differential equation is used to model the size, P (in thousands), of a species of wolves in time, t (in years) in a given habitat.

- (a) Biologists observe that if the population of wolves is “too small”, adults run the risk of being unable to find a mate, resulting in a decrease to the population.
- (i) Determine this “sparsity” value and explain how the model accounts for this observation. **[1]**
 - (ii) State the maximum population that the resources in the habitat can support. **[1]**
 - (iii) Draw a phase line diagram and comment on the stability of the equilibrium values. **[3]**
 - (iv) Sketch possible solution curves for P as a function of t . **[2]**
- (b) You are given that the population is now 2000.
- (i) Copy and complete the table showing the use of the improved Euler method with step size 0.5 to estimate the population in a year’s time. **[4]**

t	P	$\frac{1}{10} P \left(1 - \frac{P}{10} \right) (P - 1)$	$\tilde{P}$	$\frac{\Delta P}{\Delta t}$
0	2	0.16	2.08	0.16896
0.5	2.08448			
1				

- (ii) How could the accuracy of the numerical method in part (b)(i) be improved? **[1]**
- (c) Suppose the wolves are being hunted at a fixed rate E (in thousands per year).
- (i) Write down the new model for the population. **[1]**
 - (ii) Sketch a bifurcation diagram for the parameter E . Find the bifurcation value of E . **[3]**

[END OF PAPER]